AI ACADEMY · LEVEL 2 · GRADE 7 · WEEK 33 · STUDENT LESSON

Build the Recommender

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

Before you start

Week 23: score only candidates satisfying hard constraints.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Implement a deterministic recommender. Try the recall before opening notes.

Your learning target

I can implement a **deterministic recommender**.

I can handle ties and no eligible options.

I can explain recommendations with their limits.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Project Problem and Success Test. Give an example and a mistake that the idea helps you avoid.

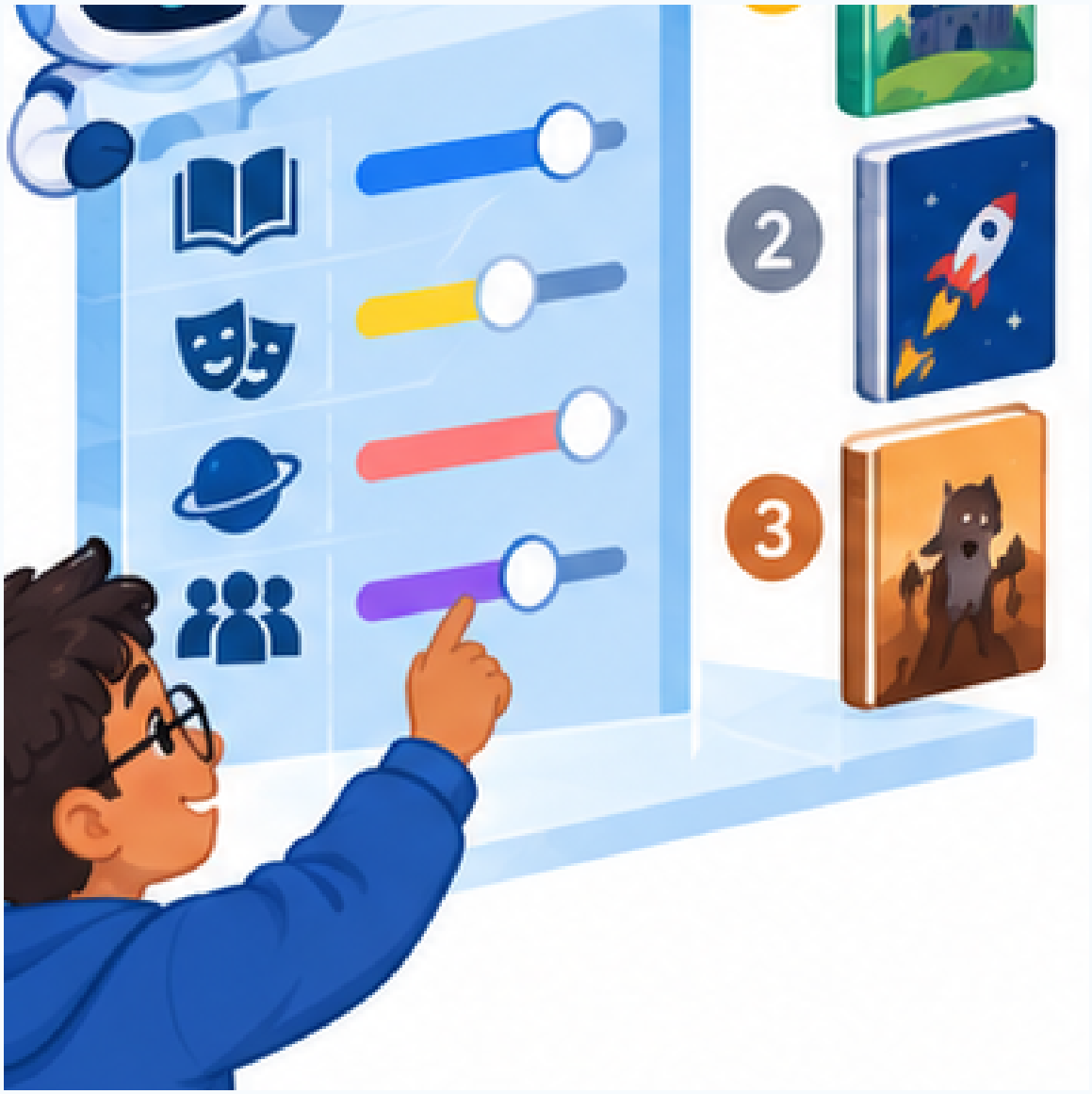
Without opening the old lesson, explain one key idea from Agents and Tools. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Predictive vs Generative. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your **explanation** that the **evidence** shows was wrong.

Week 33: Build the **Recommender**

A transparent **baseline** makes later improvements understandable.



THINKER MISSION

Implement a scoring table and produce ranked **output**s.

Thinker name and date:

Write or draw your reasoning and evidence.

Readiness for the connected explanation

An option violates a hard duration limit but has the best score. Should it win?

Readiness for Building the data pipeline

What are the steps between raw records and a recommendation?

My readiness explanation

This week's end goal

By the end, I can implement **deterministic** filtering, scoring and ties for a complete catalogue case.

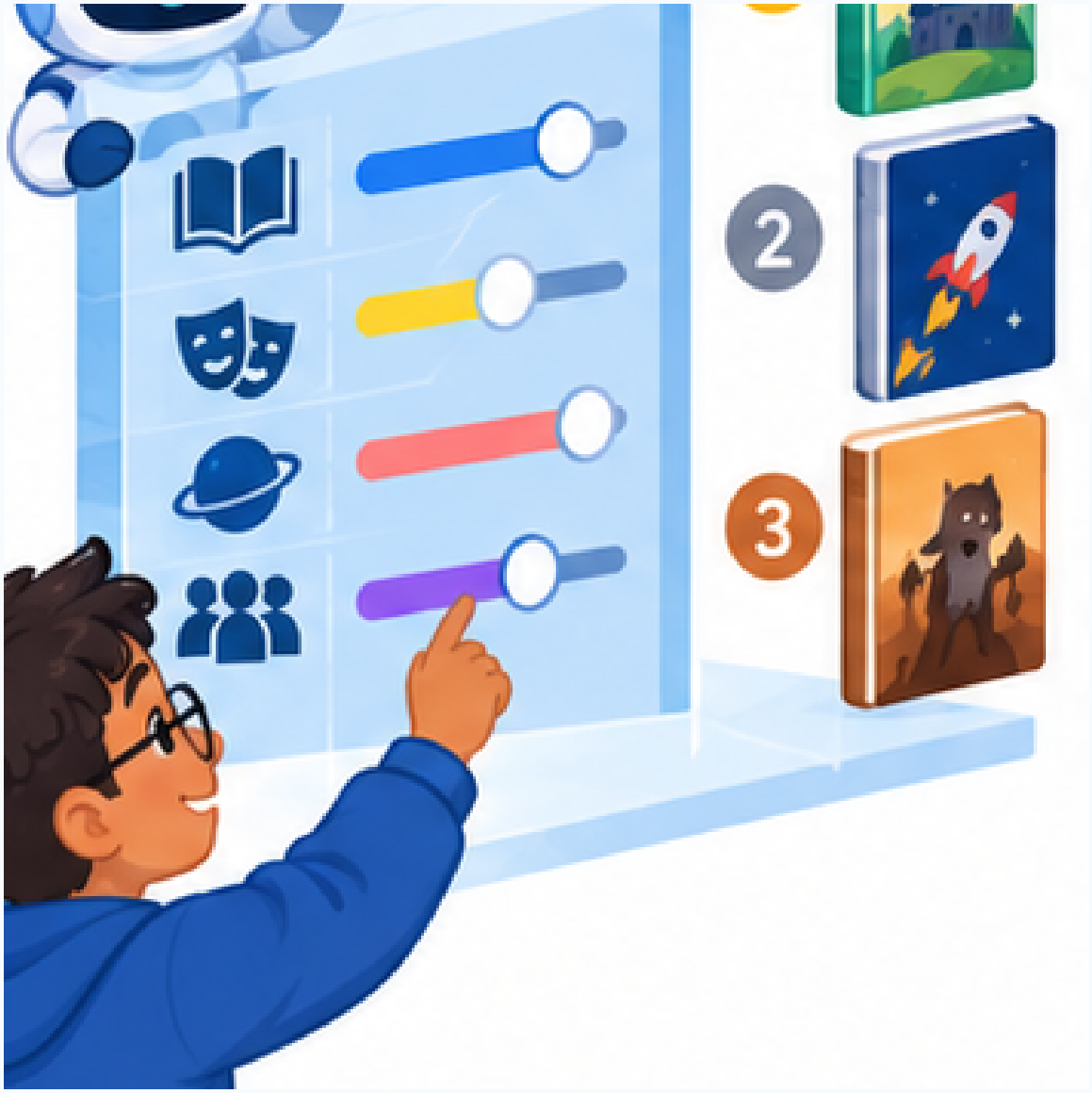
Notice the evidence

> Investigate the supplied Build the Recommender case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



Start with this puzzle

MYSTERY

The learners must use build the **recommender** to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

Write or draw your reasoning and evidence.

My **prediction** and the clue supporting it:

Write or draw your reasoning and evidence.

A closer explanation

Build a fictional activity **recommender** using last week's score. Require duration at most a supplied limit. On a tie choose the alphabetically first name.

This makes the outcome reproducible rather than dependent on **input** order. Assume unique names and validated numeric fields.

Apply the connected idea: Building the data pipeline

Build the validation **stage** of the planned chooser first. A valid record has nonempty text name, integer minutes 0 through 60 and integer fun and learning ratings 0 through 5.

The batch **stage** must separately check name uniqueness. Only accepted records may reach the Week 24 selection function.

STUDENT LESSON · 3 OF 18

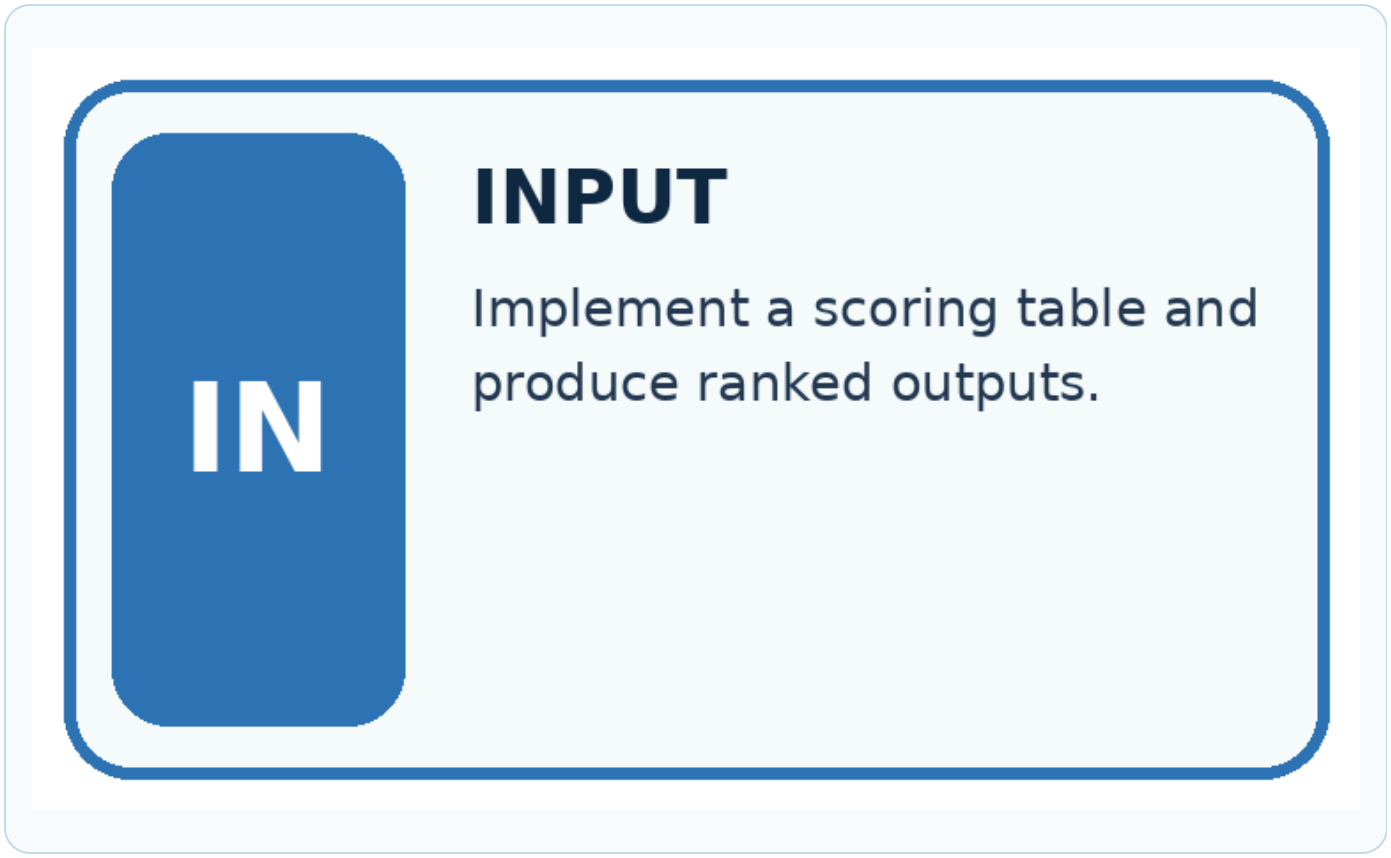
See how it works

> Follow the mechanism for Build the Recommender. Connect each labelled input, deciding step and result.

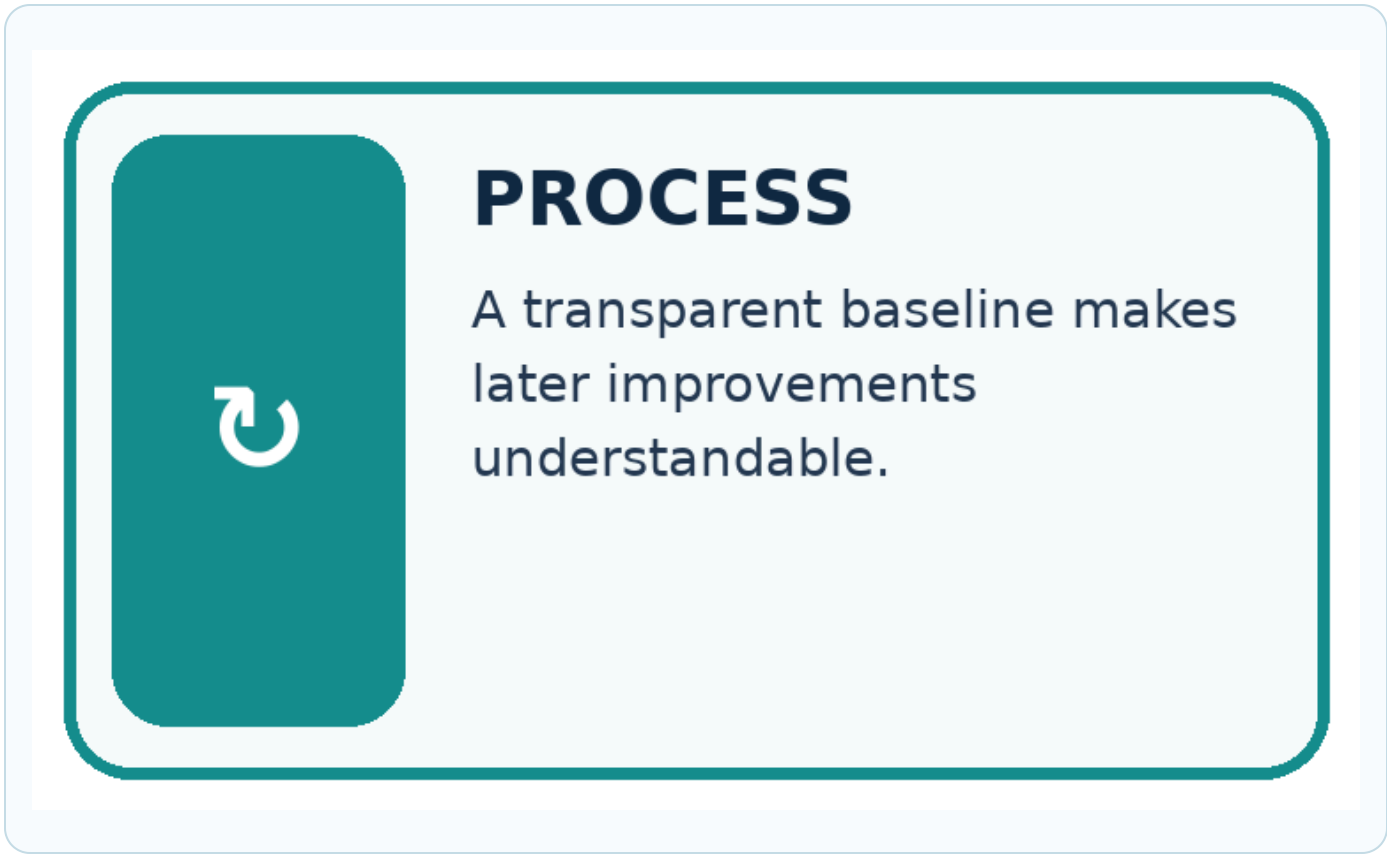
VISUAL MECHANISM

Input → **Process** → **Output**

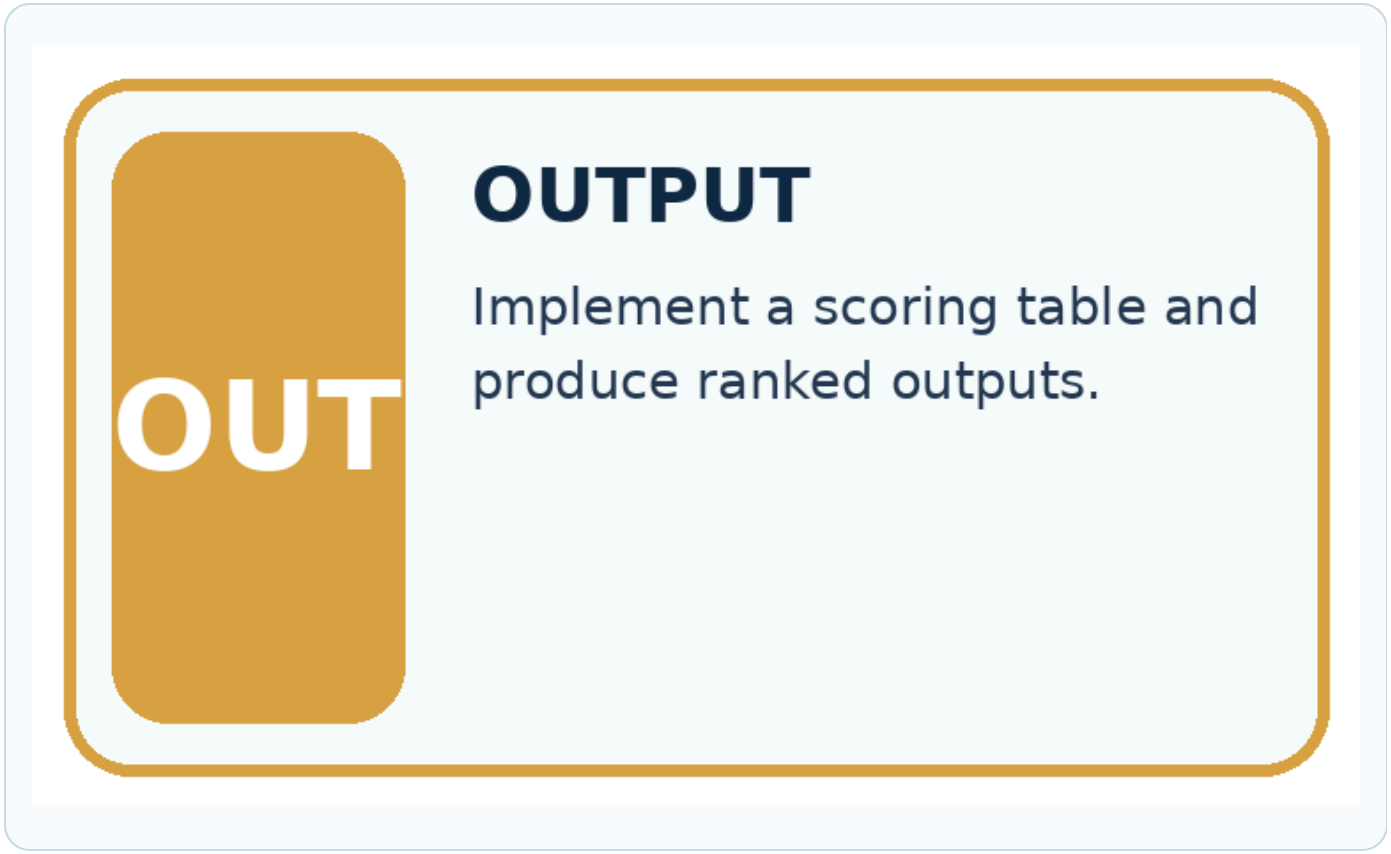
Trace the information. At each **stage**, ask what changed and how you could test it.


IN
INPUT
Implement a scoring table and produce ranked outputs.

↓


PROCESS
A transparent baseline makes later improvements understandable.

↓



Which **stage** could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

```
def recommend(items, limit):
    best = None
    best_score = None
    for item in items:
        if item["minutes"] <= limit:
            score = 2 * item["fun"] + item["learning"]
            if (best is None or score > best_score
                or (score == best_score and item["name"] < best)):
```

```
                best = item["name"]
                best_score = score
    return best

items = [{"name": "B", "fun": 3, "learning": 2, "minutes": 10},
         {"name": "A", "fun": 2, "learning": 4, "minutes": 9}]
print(recommend(items, 10))
```

Follow the evidence

STEP OR CASE	EVIDENCE	MEANING
B	Eligible score 8	Temporary best B
A	Eligible score 8	Alphabetical tie gives A

Read each row from left to right and connect the evidence to the stated rule.

The function prints nothing; the caller prints A. best starts as None to mean no eligible option found. Python short-circuits or: when best is None, later comparisons in that condition need not run.

After selecting an item, best_score is numeric. An empty list or no eligible item returns None.

Trace the best-so-far record

Write a trace table with item, eligible, score and best-so-far. In the worked list, B qualifies and scores 8, so B becomes the current best. A also qualifies and scores 8.

The scores tie, so the written alphabetical rule changes the best to A. Starting with A gives the same final answer. If no item qualifies, best stays None; do not invent a winner or quietly relax the limit.

Apply the connected idea: Building the data pipeline

```
def valid_item(row):
    if type(row) is not dict:
```

```
        return False
    if type(row.get("name")) is not str or not row["name"]:
        return False
    for key, maximum in [("minutes", 60), ("fun", 5), ("learning", 5)]:
        value = row.get(key)
        if type(value) is not int or not 0 <= value <= maximum:
            return False
    return True

print(valid_item({"name": "A", "minutes": 20, "fun": 4, "learning": 2}))
print(valid_item({"name": "B", "minutes": 10, "fun": "4", "learning": 2}))
```

Follow the evidence

STEP OR CASE	EVIDENCE	MEANING
Raw rows	Check fields and types	Accepted or rejected
Accepted rows	Check unique names	No ambiguous identifiers
Unique valid rows	Apply Week 24 selector	Suggestion plus counts

Read each row from left to right and connect the evidence to the stated rule.

The worked calls print True then False. Short-circuit checks keep absent or wrongly typed fields away from later operations.

This Boolean helper does not explain the reason; add reason codes in the reporting **stage** or use a validation result record. Do not describe a range failure as a no-match outcome.

Trace the hand-off between stages

Think of validation and selection as two desks. At the first desk, each record is checked for required fields, exact types and allowed ranges; a batch check also checks unique names.

Only accepted records go to the second desk, where the time limit and scoring policy are applied. A record can be valid yet too long to qualify.

Report rejected and ineligible separately so a reader knows which decision removed an item.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

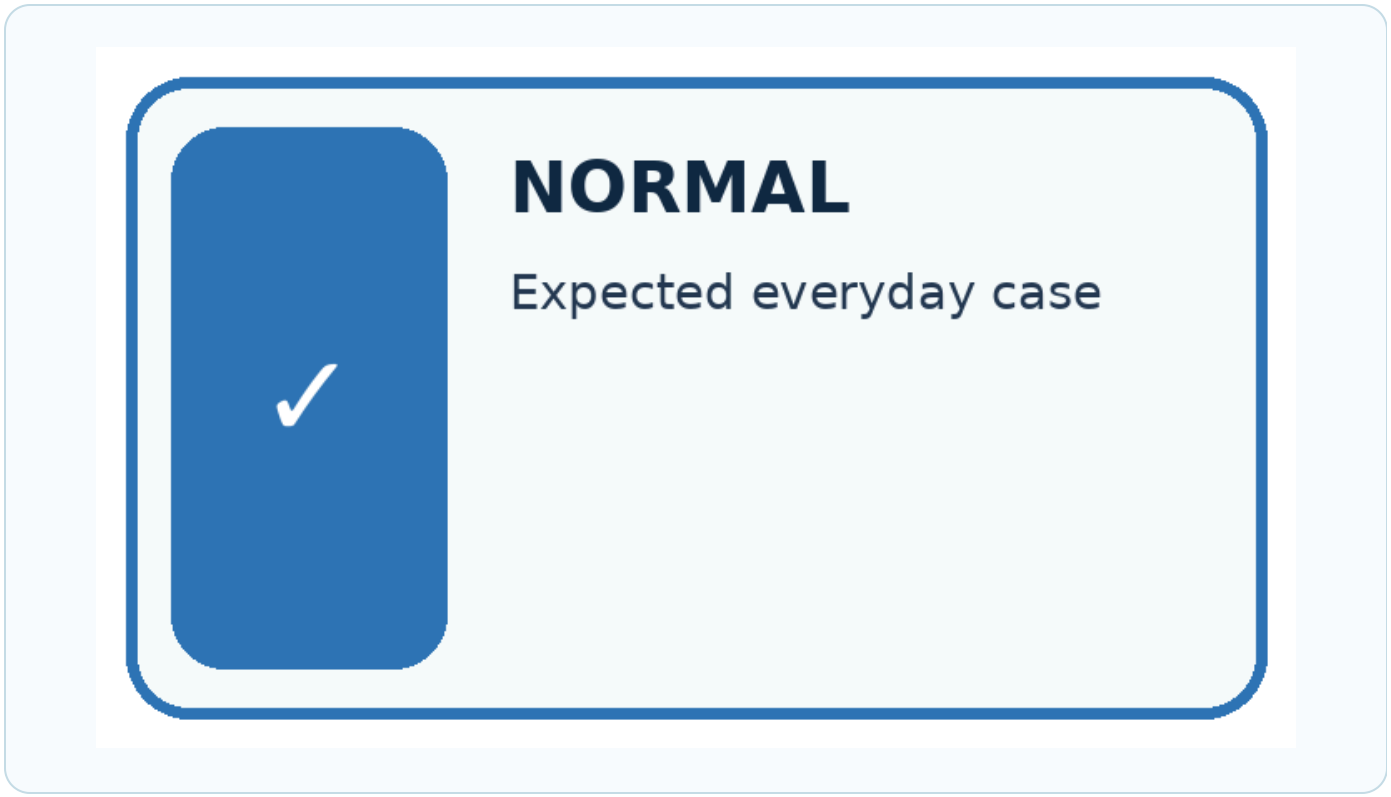
> Compare the supplied cases for Build the Recommender. Identify the detail that changes the conclusion.

The case to investigate

CASE GALLERY

Four Tests, Four Kinds of **Evidence**

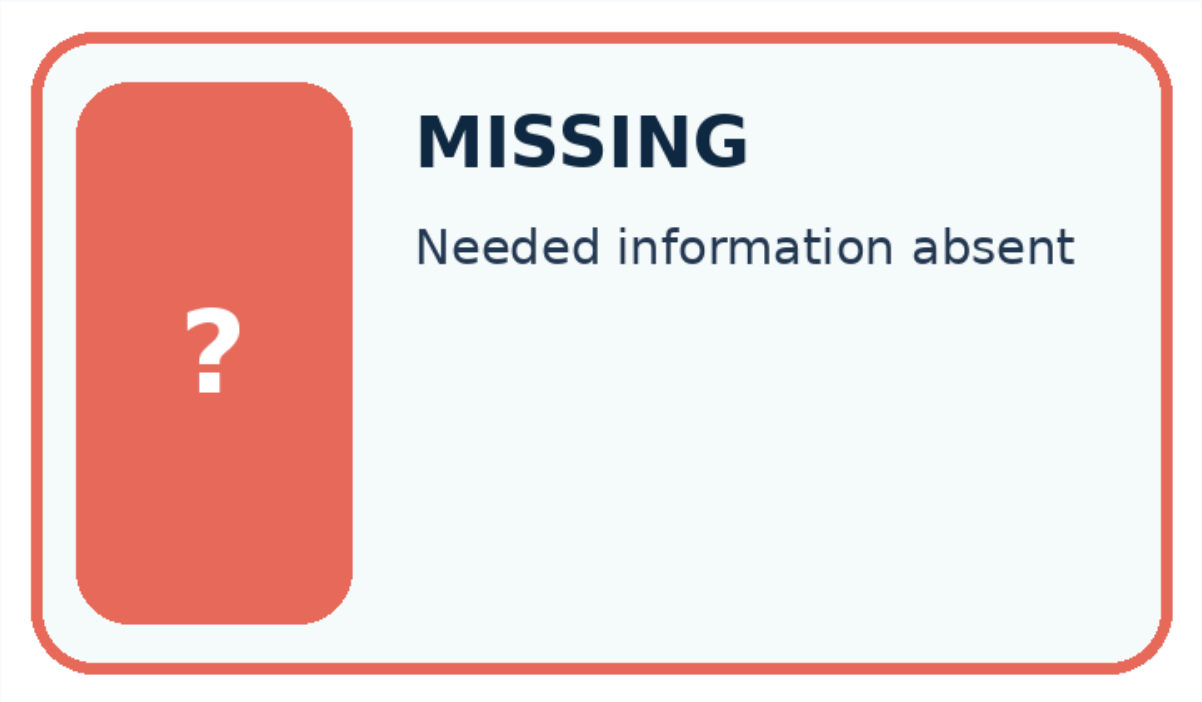
A system is not understood until it survives more than one comfortable example.


NORMAL
Expected everyday case


BOUNDARY
Near a decision edge

The normal case

NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.





Check missing information

MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Change one thing in the pictured case

Change one thing

Change: Change D’s pages to 91, leaving every other field unchanged.

Predicted result

Prediction: D becomes ineligible, so A wins with score 2.

Why the result follows

Reason: The strict excess over the 90-page maximum changes filtering before ranking.

STUDENT LESSON · 5 OF 18

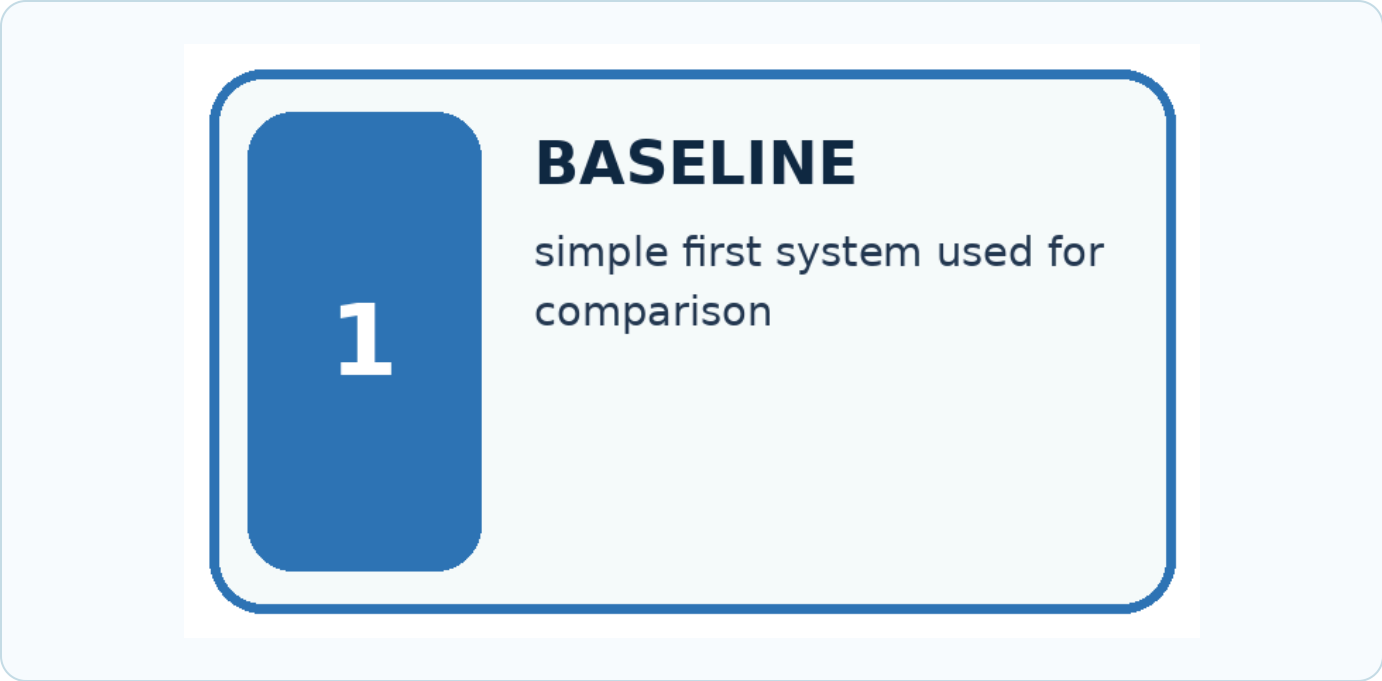
Words that unlock the idea

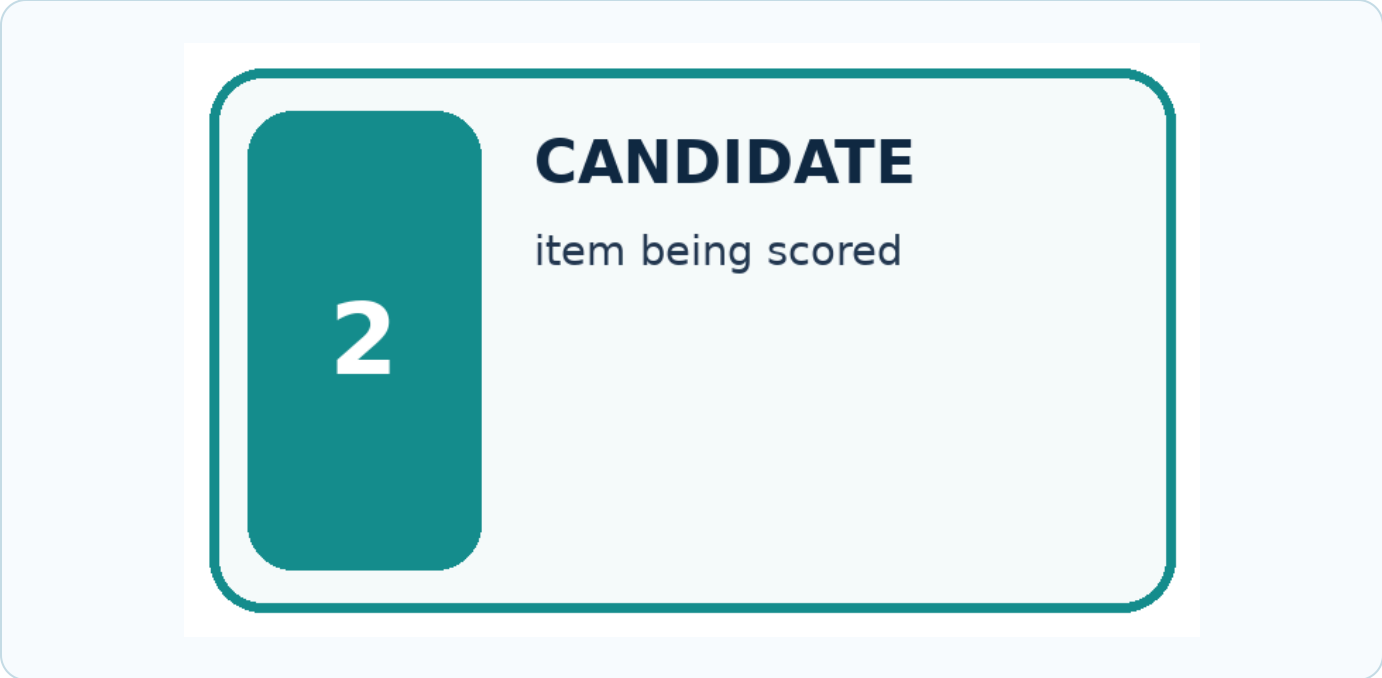
> Use the vocabulary of Build the Recommender to explain a concrete decision, rather than only memorising definitions.

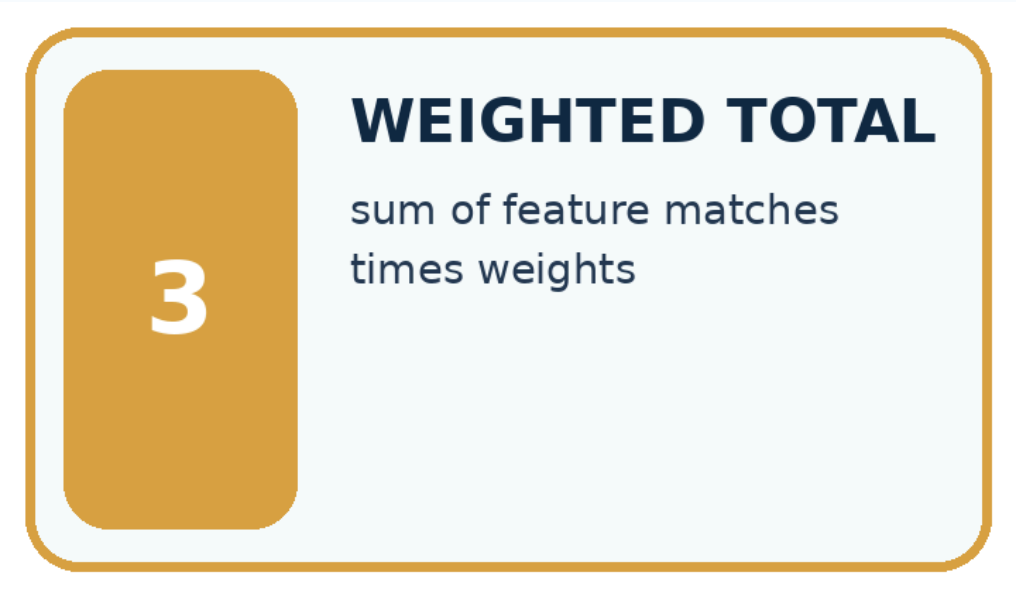
VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.


1
BASELINE
simple first system used for comparison


2
CANDIDATE
item being scored





Word	My own example	Not this / boundary
baseline	A simple reference method used to judge whether a more complex method adds value.	
candidate	An item or proposed answer being considered before a final selection.	
weighted total	A sum formed after multiplying each contribution by its assigned weight.	

Word	My own example	Not this / boundary
explanation	An account connecting a result to the mechanism, calculation or evidence that supports it.	

Words used in the closer explanation

Word	Meaning
Recommender	A procedure that suggests options using a stated policy
Tie	Two options have the same comparison score
Deterministic	The same inputs and rules produce the same result

Terms for Building the data pipeline

Term	Meaning
Stage	One part of a processing pipeline
Validation function	A function that decides whether an input meets a schema
Separation of concerns	Keeping different responsibilities in distinct parts
Rejection reason	An explanation of why a record was not accepted

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STUDENT LESSON · 6 OF 18

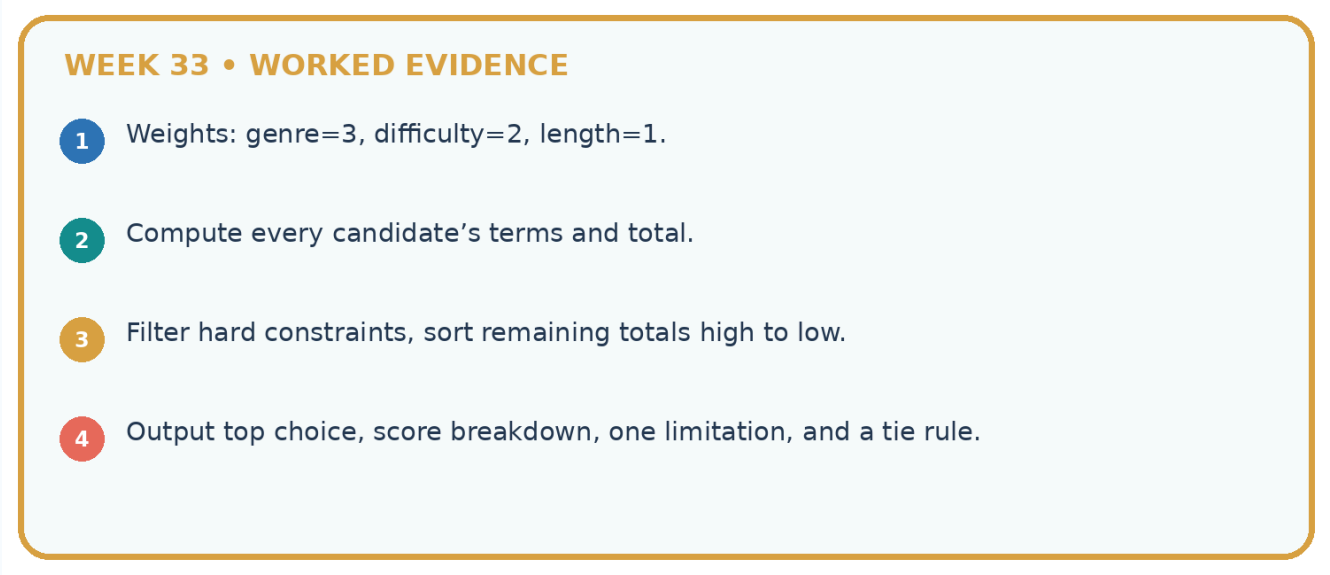
Follow a worked example

> Follow the worked trace for Build the Recommender, predict a step, then compare with its explanation.

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.



1. Weights: genre=3, difficulty=2, length=1.

2. Compute every **candidate**’s terms and total.

3. Filter hard constraints, sort remaining totals high to low.

4. **Output** top choice, score breakdown, one limitation, and a tie rule.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Predict the result if the list order is reversed, and if the limit is 8.

Apply the connected idea: Building the data pipeline

Predict validity for minutes 0, fun 0, learning 5 with name A, then for minutes 61.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Build the Recommender

A fictional book chooser filters a request before comparing preference scores.

The facts we will use

Supplied facts

Request: animal subject, at most 90 pages. Score 2 for illustrated and 1 for comic. Tie chooses smaller ID.

Supplied facts

A animal90 illustrated prose; B animal70 plain comic; C space60 illustrated comic; D animal80 illustrated comic.

AI ACADEMY / AI-2/33

Filtering before scoring

	Pages	Preference score	Status
A	90	2	eligible
B	70	1	eligible
C	60	3	subject fails
D	80	3	eligible

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-2/33

Build the Recommender

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Filter

A,B,D meet subject and page requirements; C fails subject.

2. Compute scores

A=2; B=1; D=3.

3. Choose best

Return D with both preference matches recorded.

4. Deterministic output

Reordering the input rows must not change D's unique winning score.

CHANGE ONE THING

Change D's pages to 91, leaving every other field unchanged.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Filter	A,B,D meet subject and page requirements; C fails subject.
2. Compute scores	A=2; B=1; D=3.
3. Choose best	Return D with both preference matches recorded.
4. Deterministic output	Reordering the input rows must not change D's unique winning score.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Eligibility and ranking answer different questions. C has a high preference score but is not an animal book, so it must stay excluded.

The implementation needs a no-match result and explicit tie policy even when this example has a unique winner. Save rejected reasons for an **explanation**, not just the final ID.

What this example does not establish

Keep the stated limit

Toy preference weights do not imply shorter or illustrated books are universally better for all learners.

Keep the stated limit

Read the labelled picture: Explain why B does not win after D is excluded even though it has fewer pages.

Keep the stated limit

Change one thing: Change D's pages to 91, leaving every other field unchanged. Predict the result before checking the **explanation**. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Build the Recommender claim and explain a repair using the supplied evidence.

Find the first error

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.

!

COMPARE • DIAGNOSE

Weak idea: A recommendation is transparent if it displays only the final score.
Strong move: locate evidence, change one cause, and retest.

Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

The recommendation follows a written scoring policy. It does not learn preferences, prove learning benefit or authorize an action. Display the chosen rule and let a person review whether an option is suitable.

> **Write a one-sentence explanation of X winning without calling it objectively best.**

Apply the connected idea: Building the data pipeline

Each **stage** should have a clear **input** and **output** contract. A program can be correct at one **stage** but wrong at their connection, such as passing rejected rows into selection. Keep acceptance and rejection counts in the report.

> **Two valid rows have the same name but different ratings. Should individual validation alone permit selection?**

STUDENT LESSON · 8 OF 18

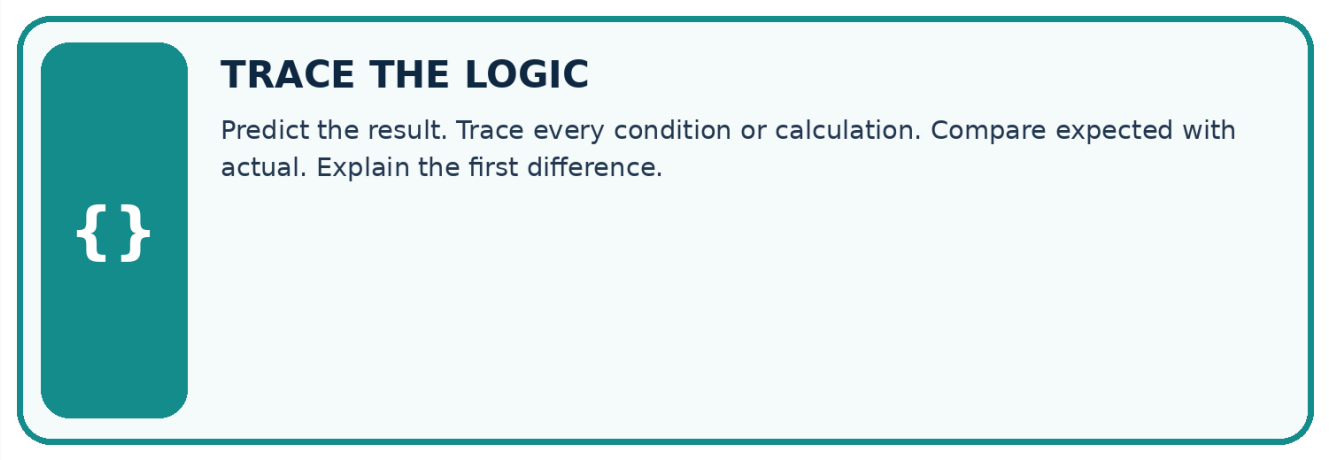
Practise together

> Practise Build the Recommender with support, then explain the deciding step in your own words.

LOGIC STUDIO

Trace Before You Run

Block code, pseudocode, tables, and diagrams are different ways to expose the same reasoning.



SET **evidence**_record PREDICT expected_result RUN one normal case and one boundary case COMPARE actual_result with expected_result IF different: locate first cause; change one factor; RETEST EXPLAIN result, limitation, and next safe step

> Trace one **input**. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

Updating best on score >= best_score makes the last tied item win, so reordering can change results. Use the explicit name comparison for ties. Test both orders and a no-match case after the repair.

Apply the connected idea: Building the data pipeline

Calling the selector on raw data can crash or produce a result from invalid ratings. Route data through validation first.

Duplicate names require a batch check even when each row individually passes; block the ambiguous batch for review instead of arbitrarily keeping one.

Practise with less help: Handle no match

Use workbook W2 for the connected example “Implementing a **recommender**”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own **explanation**.

Check your reasoning

Use the worked case for Implementing a **recommender**. Proposed **explanation**: “A successful run proves the conclusion”. Decide whether this **explanation** is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Practise the connected topic: Building the data pipeline

Use its named workbook W2. Identify the given **input**s, try the middle steps with less help, then explain your trace. Check this claim before group discussion: A successful run proves the conclusion

Explain the picture

Explain why B does not win after D is excluded even though it has fewer pages.

> **My explanation using a pictured step and the given facts**

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

Run the investigation

> Investigate Build the Recommender. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



Build the editable score table, hand-check one row, run two profiles, and compare system ranking with independent recalculation.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
>
> **Prediction:** _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: boundary

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: missing

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: fresh transfer

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

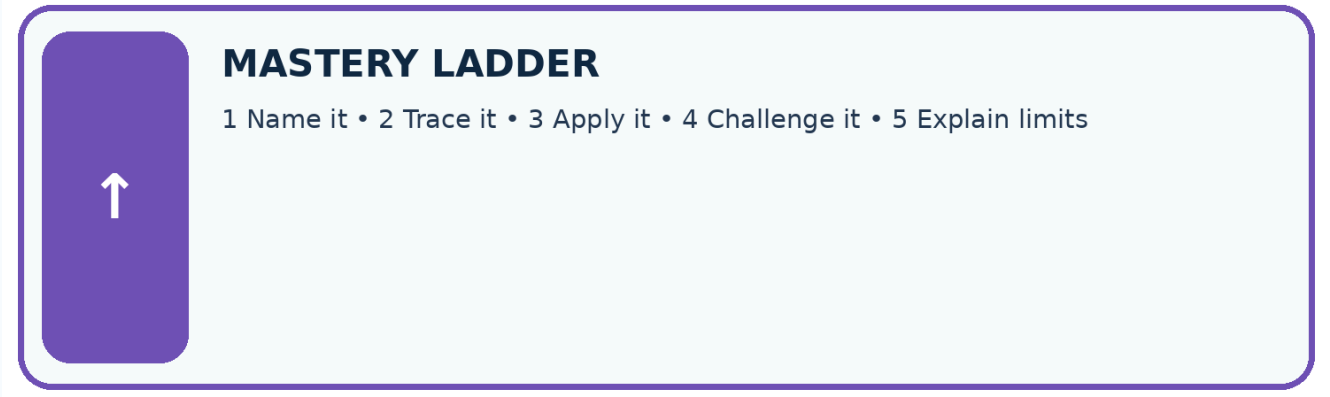
Build your understanding

> Move from recall to tracing, diagnosis and transfer in Build the Recommender. Explain why each step is justified.

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



1 • NAME: Define the concept without repeating the textbook.

2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.

3 • APPLY: Solve a different example independently.

4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest **evidence** of independent understanding:

> **Write or draw your reasoning and evidence.**

STUDENT LESSON · 11 OF 18

Connect your project

> Apply this week's idea to Transparent Book Recommender. Keep an annotated decision and the evidence supporting it.

CUMULATIVE PROJECT

Transparent Book **Recommender**

Every week adds one inspectable piece. The system must show how it reached a result.

TRANSPARENT BOOK RECOMMENDER

This is the complete baseline: inputs, scoring, filter, ranking, explanation, and abstain rule.

THIS WEEK'S CHECKPOINT This is the complete **baseline**: **input**s, scoring, filter, ranking, **explanation**, and abstain rule.

Trace the system

SYSTEM MAP Reader need → allowed **input**s → transparent rule or score → ranked **candidate**s → **explanation** → test record → **human decision**

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under **human control**?

Write or draw your reasoning and evidence.

Responsible design

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse a ranking with a fact about a person.

Evidence for your project

Save the **recommender** code, contract and this full trace; test the same records in different **input** orders.

STUDENT LESSON · 12 OF 18

Show what you know

Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Independent evidence

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.

✓

INDEPENDENT PROOF

Implement a scoring table and produce ranked outputs.

Try a more demanding case

CHALLENGE Implement a scoring table and produce ranked **output**s.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

2. My **evidence** that it works (include a boundary or missing-data case):

Write or draw your reasoning and evidence.

3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

Check your explanation

SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

STUDENT LESSON · 13 OF 18

Study a complete case

Revisit the complete Build the Recommender case for supported practice. Repeated examples do not become fresh assessments.

Week 33 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

A has flags (1,1), B=(1,0), C=(0,1); score=2g+l.

The question

Compute the winner and **evidence** to display.

Worked explanation

A wins at 3; B=2; C=1. Show flags, formula, scores and **explanation**; declare tie policy.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Build and expose the scoring calculation

Evidence or stage	Value or result
A, genre and length	3
B, genre only	2
C, length only	1

A has flags (1,1), B=(1,0), C=(0,1); score=2g+l.

Compute the winner and **evidence** to display.

A wins at 3; B=2; C=1. Show flags, formula, scores and **explanation**; declare tie policy.

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Changed independent case

Request: mystery, at most 100 pages. Catalogue: D mystery110 illustrated; E mystery80 plain; F science80 illustrated; G mystery95 illustrated; H mystery90 illustrated.

Filter first; score 3 for mystery plus 2 for illustrations; break ties by fewer pages.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 33 Independent application

Close the worked **explanation**. Apply the idea to this different question without copying.

> A reader requests space and at most 60 pages. Book A is space, 90 pages; B is animals, 40 pages. What should the prototype return if both conditions are required?

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

Write or draw your reasoning and evidence.

Connect to your project

This is the complete **baseline**: **input**s, scoring, filter, ranking, **explanation**, and abstain rule.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

Catalogue: A mystery/90/illustrated, B mystery/140/not illustrated, C science/80/illustrated. Request mystery, maximum 100 pages. Rank with 2 genre points and 1 illustration point after filtering.

Show why other books are excluded.

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Build the Recommender, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: A fictional book chooser filters a request before comparing preference scores.

By the end, I can	How I will show it
Implement a deterministic recommender	Explain the deciding step, show a trace or test, and state one limit.
Handle ties and no eligible options	Explain the deciding step, show a trace or test, and state one limit.

By the end, I can	How I will show it
Explain recommendations with their limits	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Build the Recommender without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Implement a **deterministic recommender**

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional book chooser filters a request before comparing preference scores.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Reordering the **input** rows must not change D’s unique winning score.

1. Choose best

2. Filter

3. **Deterministic output**

4. Compute scores

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the **stage** that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported **explanation**: **Deterministic output** Reordering the **input** rows must not change D's unique winning score. Eligibility and ranking answer different questions.

C has a high preference score but is not an animal book, so it must stay excluded. The implementation needs a no-match result and explicit tie policy even when this example has a unique winner.

Save rejected reasons for an **explanation**, not just the final ID.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Change D's pages to 91, leaving every other field unchanged. Which conclusion is supported by the supplied facts?

1. D becomes ineligible, so A wins with score 2.

2. This one example guarantees correct results for every future **input**.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported **explanation**: D becomes ineligible, so A wins with score 2. The strict excess over the 90-page maximum changes filtering before ranking.

Repair a missing building block

Review Project Problem and Success Test (ai-2/32). Goal: Define a small application purpose. Short practice: Write an equality test and explain which requirement it checks.. Return to this week and explain what you changed.

Review Agents and Tools (ai-2/30). Goal: Explain the **mechanism** and terminology of state machines for bounded agents. Short practice: Can the assistant accept Trace B because its final state is done?.

Return to this week and explain what you changed.

Review Predictive vs Generative (ai-2/27). Goal: Distinguish discriminative **prediction** from generation by inspecting the requested **output**.

Short practice: Does matching the physical box in the changed table prove the generator measures boxes?. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

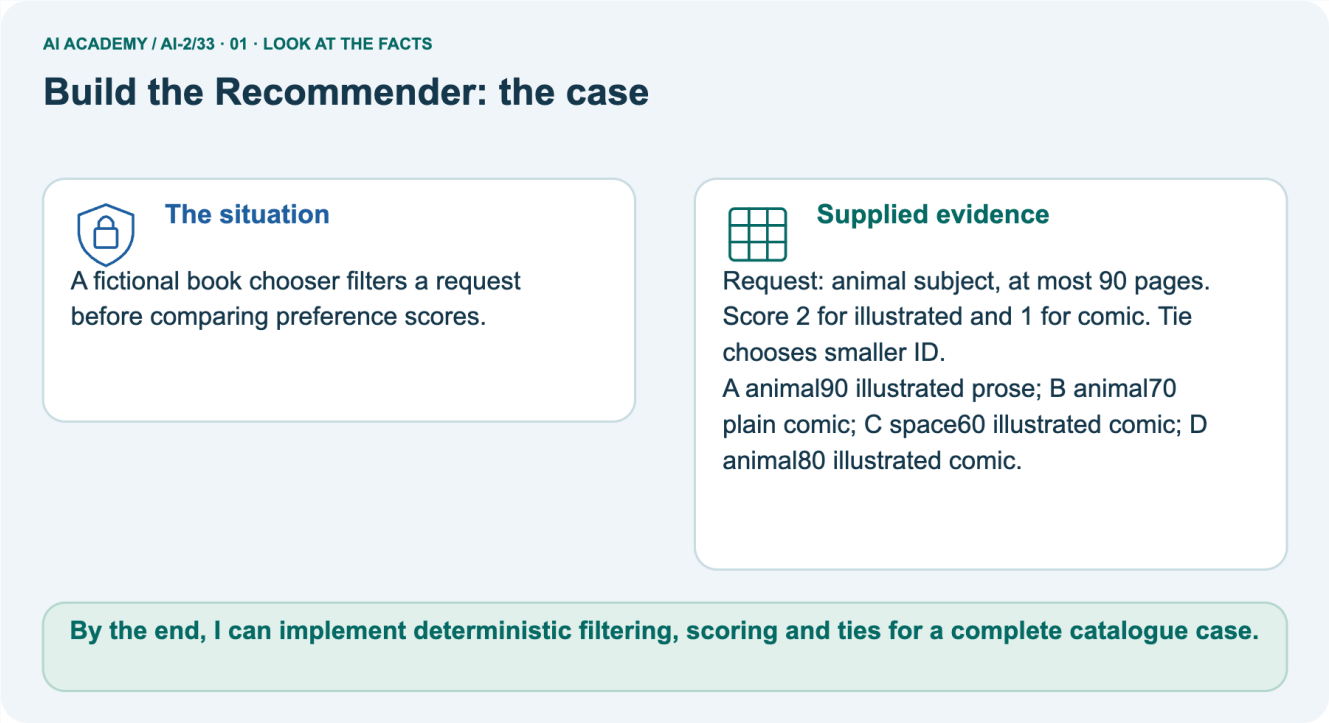
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Build the Recommender graphic using its labels, deciding evidence and relevant limitation.

01. Build the Recommender: the case



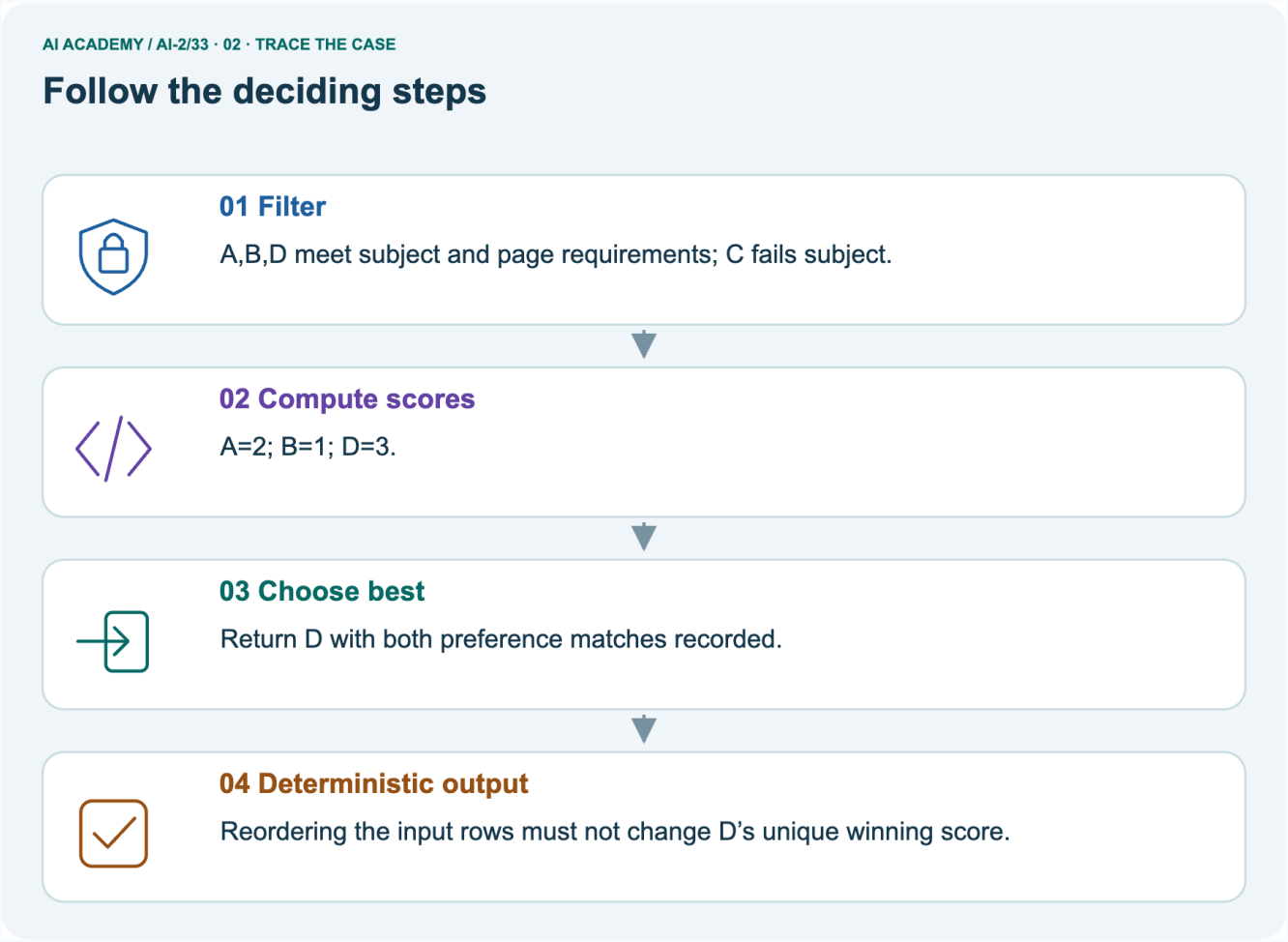
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional book chooser filters a request before comparing preference scores. Supplied facts: Request: animal subject, at most 90 pages. Score 2 for illustrated and 1 for comic. Tie chooses smaller ID.

A animal90 illustrated prose; B animal70 plain comic; C space60 illustrated comic; D animal80 illustrated comic.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

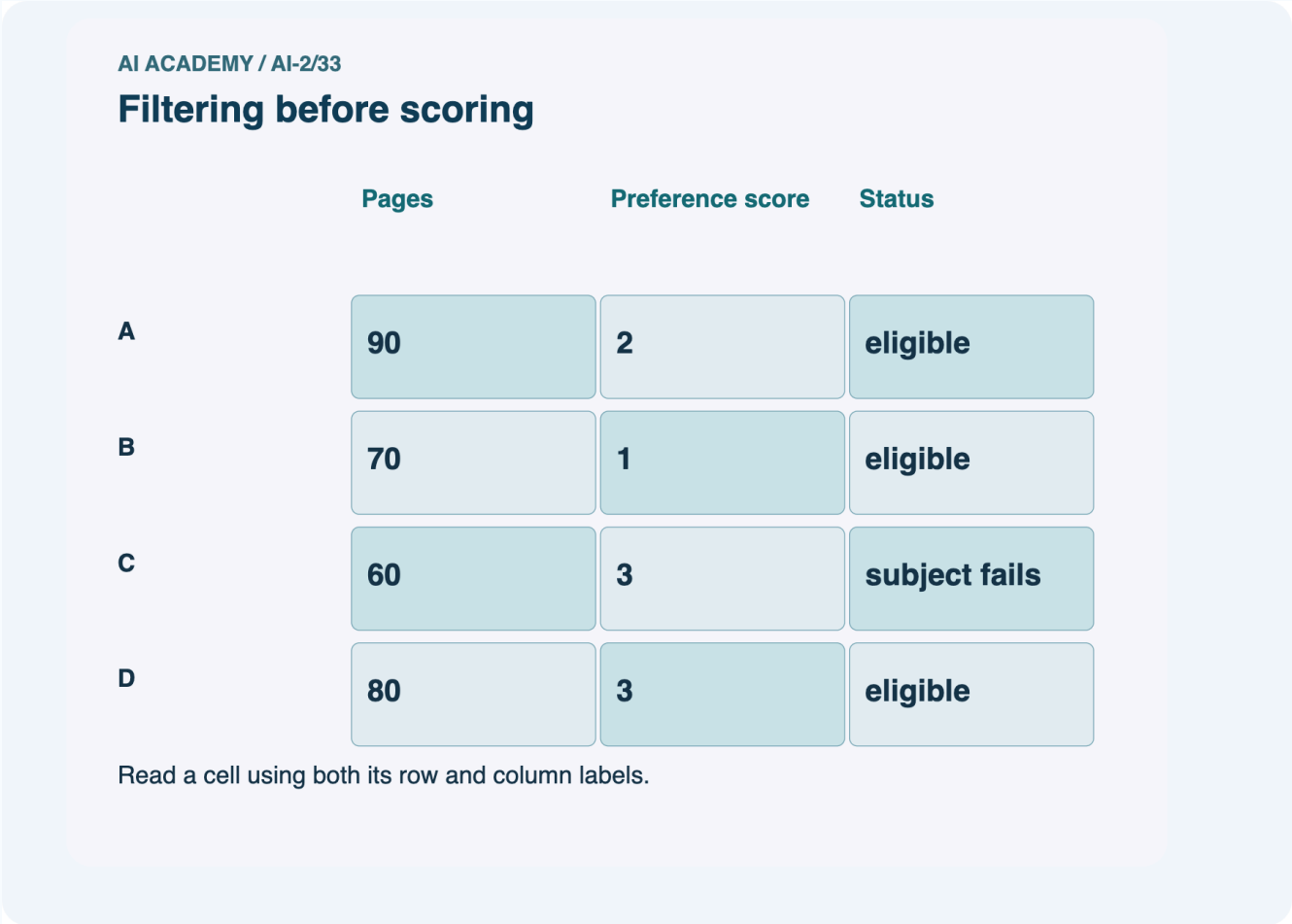


Follow the labelled picture with the matching explanation. Open the vector diagram

Filter: A,B,D meet subject and page requirements; C fails subject. → Compute scores: A=2; B=1; D=3. → Choose best: Return D with both preference matches recorded. → **Deterministic output**: Reordering the **input** rows must not change D's unique winning score.

> Point to each numbered **stage** and explain which recorded **input** it uses.

03. Filtering before scoring



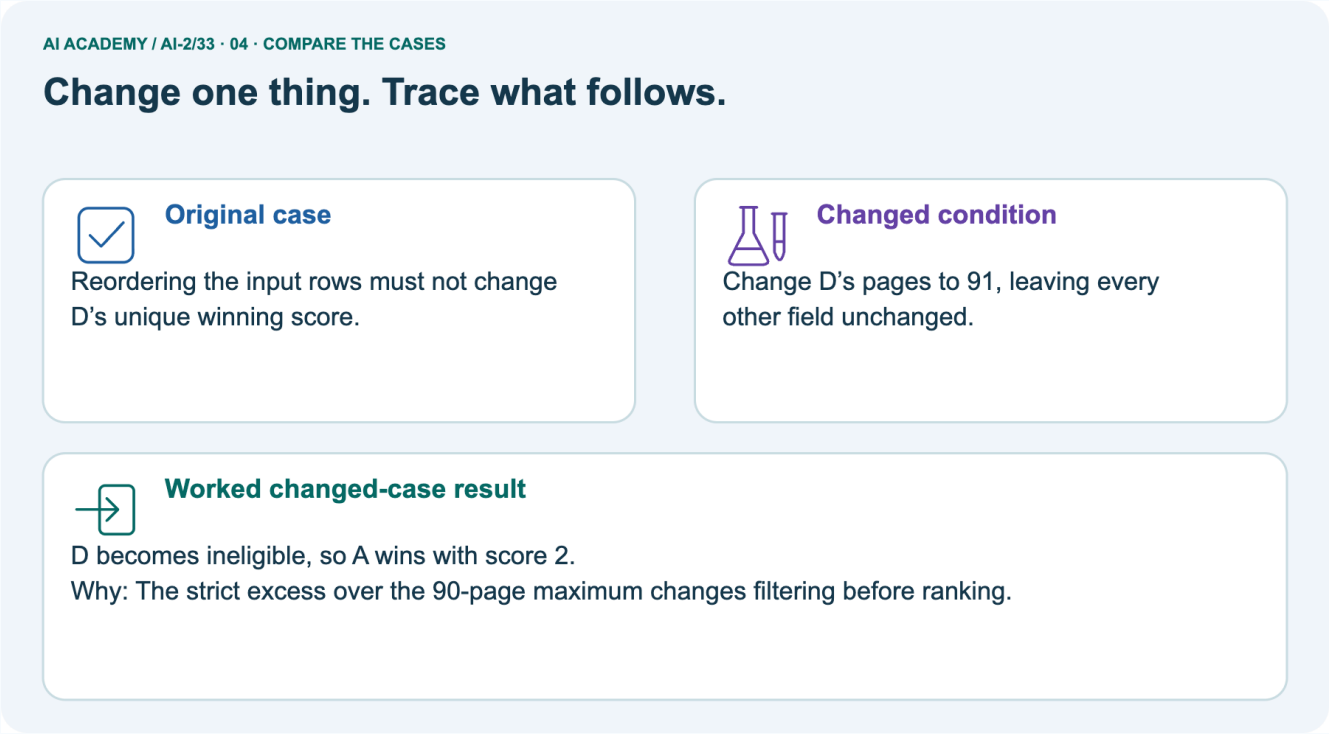
Follow the labelled picture with the matching explanation. Open the vector diagram

Filtering before scoring. A: Pages = 90, Preference score = 2, Status = eligible. B: Pages = 70, Preference score = 1, Status = eligible. C: Pages = 60, Preference score = 3, Status = subject fails.

D: Pages = 80, Preference score = 3, Status = eligible.

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



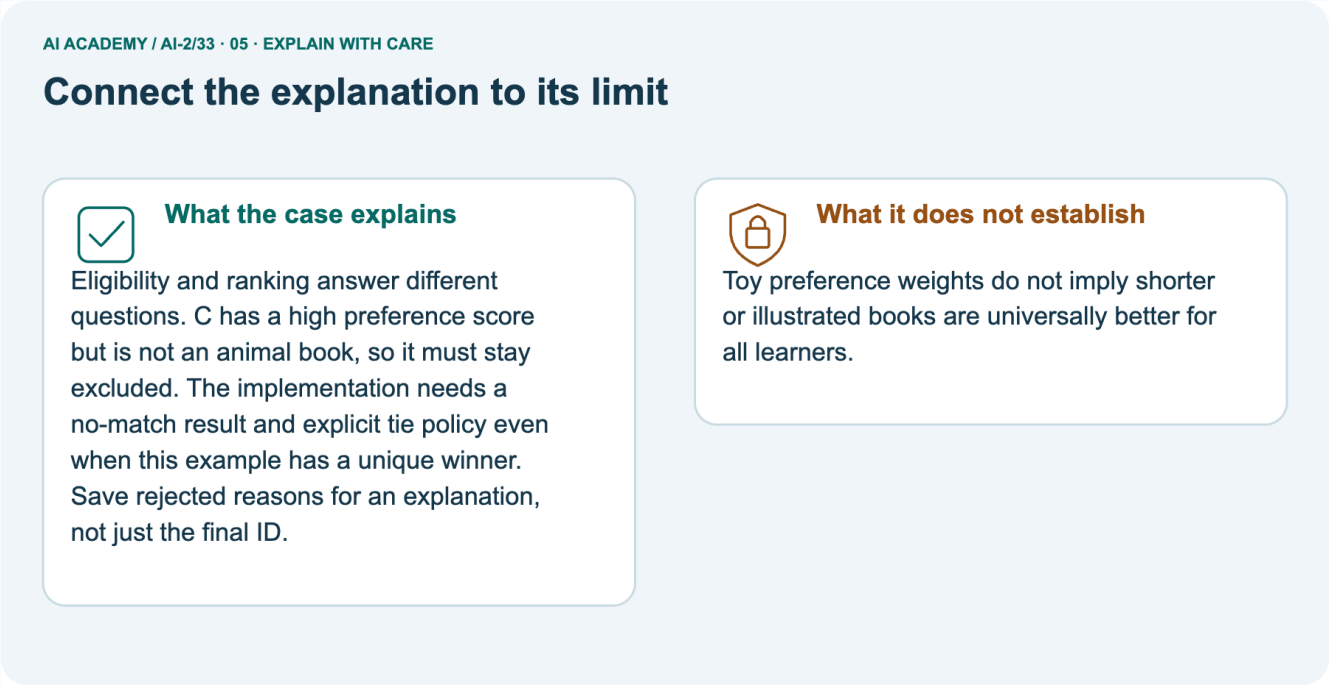
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Reordering the **input** rows must not change D's unique winning score. Changed condition: Change D's pages to 91, leaving every other field unchanged. Expected result: D becomes ineligible, so A wins with score 2.

Reason: The strict excess over the 90-page maximum changes filtering before ranking.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

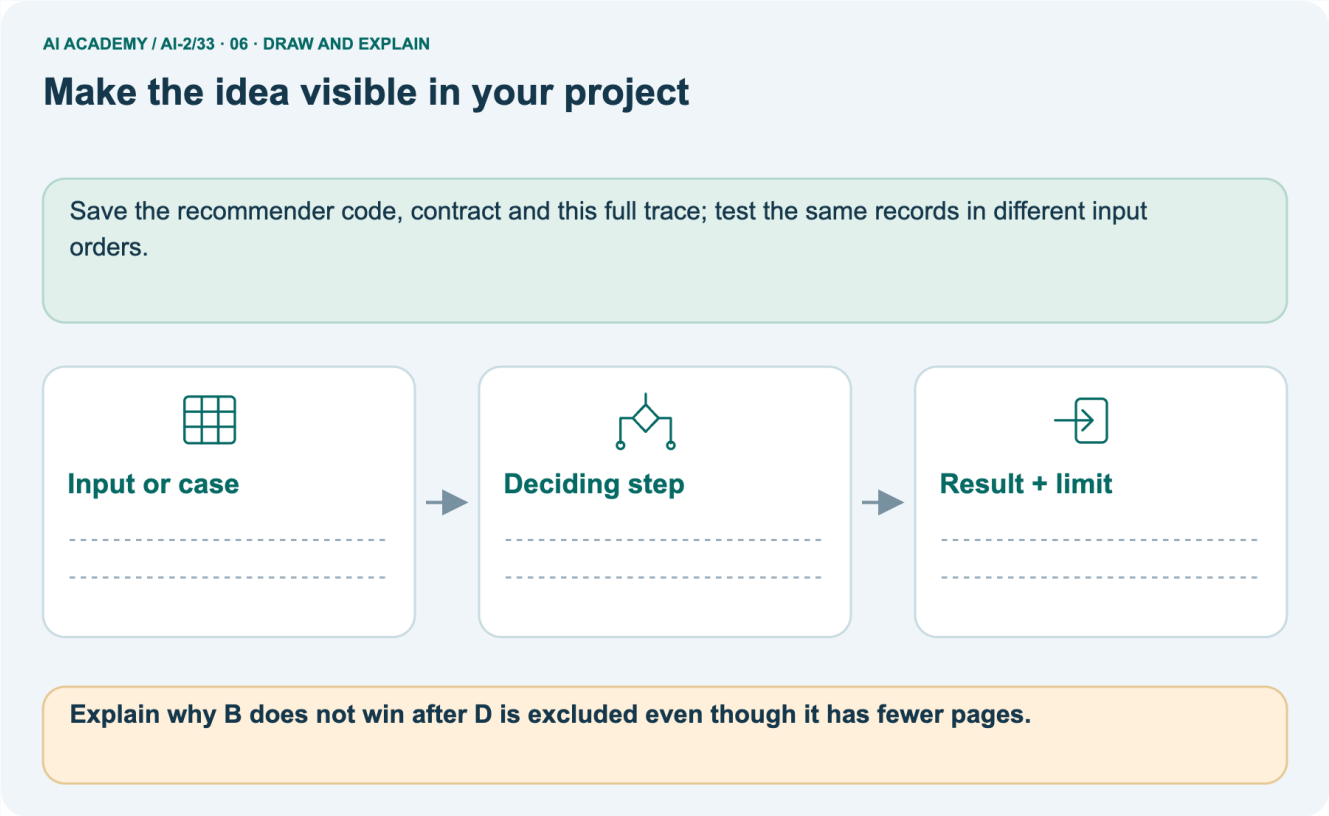
Eligibility and ranking answer different questions. C has a high preference score but is not an animal book, so it must stay excluded.

The implementation needs a no-match result and explicit tie policy even when this example has a unique winner. Save rejected reasons for an **explanation**, not just the final ID.

Limit: Toy preference weights do not imply shorter or illustrated books are universally better for all learners.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Save the **recommender** code, contract and this full trace; test the same records in different **input** orders. Draw your **input**, deciding step, result and limit.

Supported question: Explain why B does not win after D is excluded even though it has fewer pages.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.